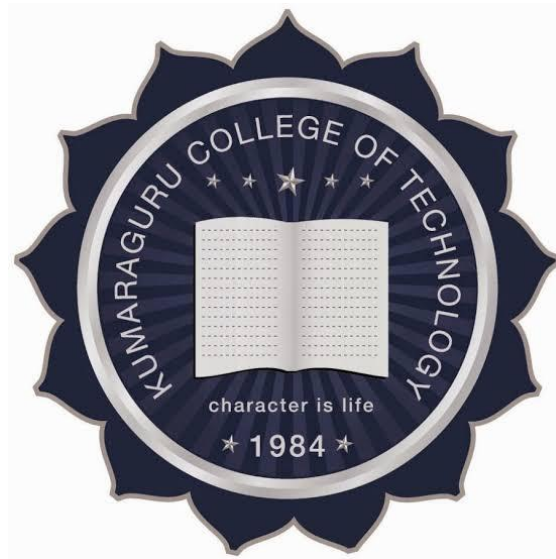


KUMARAGURU COLLEGE OF TECHNOLOGY

COIMBATORE – 641049

Department of Electronics and Communication Engineering



PROJECT REPORT

U18INI5600P – ENGINEERING CLINIC -V

Regulation 2018

NAME : SRINATH.S

ROLL NO : 19BEC167

YEAR / SEM : 3rd YEAR/ 5TH SEMESTER



KUMARAGURU COLLEGE OF TECHNOLOGY

COIMBATORE – 641049

**DEPARTMENT OF ELECTRONICS AND COMMUNICATION
ENGINEERING**

U18INI5600P – ENGINEERING CLINIC -V

Bona-fide Certificate

ROLL NO

19BEC167

Certified that this is the bona-fide record of work done by Mr. SRINATH. S
of the 5TH semester in Electronics and Communication Engineering branch during
the academic year 2021-2022

Staff in – Charge

Submitted for the University Practical Examination held on

Internal Examiner

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CAR PARKING SYSTEM using IOT

INTRODUCTION:

Car parking systems are used in parking areas to maintain a hassle-free usage of parking spaces. The system is easy and simple to design and implement.

PROBLEM STATEMENT:

All the parking areas have a specified number of parking slots, but if all the slots are occupied and another vehicle enters the area it will be no use of entering to park the vehicle as all the spaces are already occupied. This is the problem in all the parking areas.

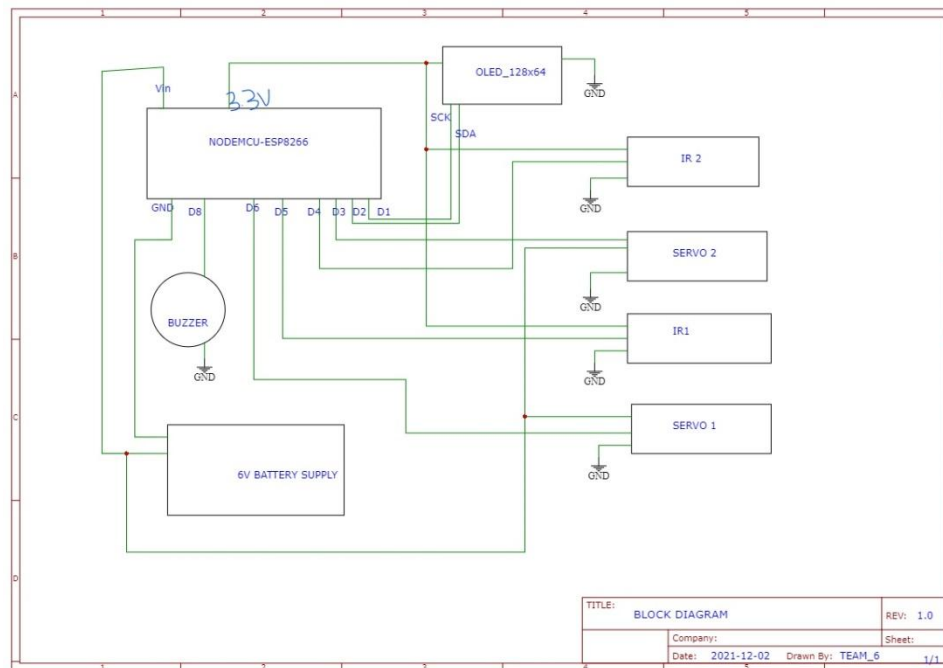
ABSTRACT:

Our IOT project is designed to solve the specified problem in the parking areas, here the ESP8266-WiFi module aka nodemcu is used as the main controlling unit, Infrared sensor modules are used to detect the vehicles movement and servo motors are used to rotate the barriers. The information will be displayed through 128*64 OLED display, and the same info will be displayed in a webpage that is programmed into the nodemcu.

COMPONENTS USED:

- I) ESP8266 WiFi module(nodemcu)
- II) Infrared sensors
- III) 9g servo motors
- IV) OLED display(128*64)
- V) 6v power supply
- VI) Buzzer
- VII) Simple switch

BLOCK DIAGRAM:



COMPONENTS DESCRIPTION:

I) NODEMCU-ESP8266:

The NodeMCU (Node MicroController Unit) is an open source software and tackle development terrain erected around an affordable SystemonaChip (SoC) called the ESP8266. The ESP8266, designed and manufactured by Espressif Systems, contains the pivotal rudiments of a computer CPU, RAM, networking (WiFi), and indeed a ultramodern operating system and SDK. That makes it an excellent choice for Internet of Effects (IoT) systems of all kinds.

Still, as a chip, the ESP8266 is also hard to pierce and use. You must solder cables, with the applicable analog voltage, to its legs for the simplest tasks similar as powering it on or transferring a keystroke to the “ computer” on the chip. You also have to program it in low position machine instructions that can be interpreted by the chip tackle. This position of integration isn't a problem using the ESP8266 as an bedded regulator chip in mass produced electronics. It's a huge burden for potterers, hackers, or scholars who want to experiment with it in their own IoT systems.

But, what about Arduino? The Arduino design created an open source tackle design and software SDK for their protean IoT regulator. Analogous to NodeMCU, the Arduino tackle is a microcontroller board with a USB connector, LED lights, and standard data legs. It also defines standard interfaces to interact with detectors or other boards. But unlike NodeMCU, the Arduino board can have different

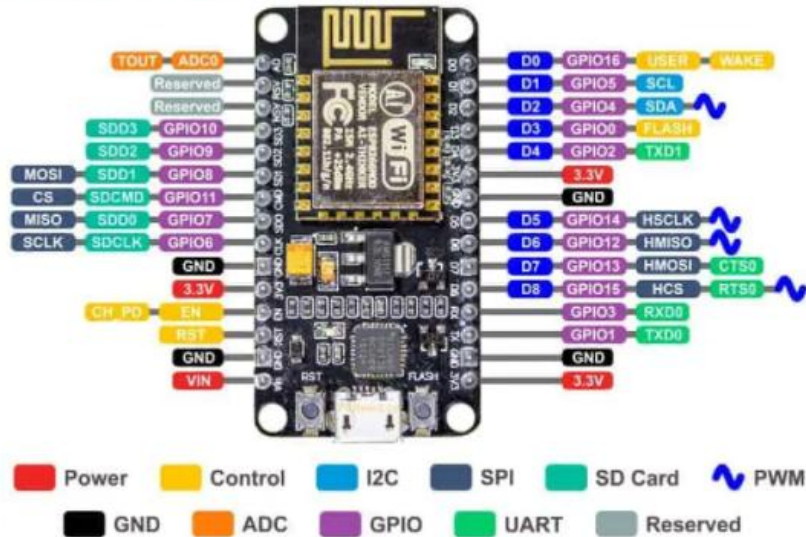
types of CPU chips (generally an ARM or Intel x86 chip) with memory chips, and a variety of programming surroundings. There's an Arduino reference design for the ESP8266 chip as well. Still, the inflexibility of Arduino also means significant variations across different merchandisers. For illustration, utmost Arduino boards don't have WiFi capabilities, and some indeed have a periodical data harborage rather of a USB port.

NodeMCU Technical Specifications

	Official NodeMCU	NodeMCU Carrier Board	LoLin NodeMCU
Microcontroller	ESP-8266 32-bit	ESP-8266 32-bit	ESP-8266 32-bit
NodeMCU Model	Amica	Amica	Clone LoLin
NodeMCU Size	49mm x 26mm	49mm x 26mm	58mm x 32mm
Carrier Board Size	n/a	102mm x 51mm	n/a
Pin Spacing	0.9" (22.86mm)	0.9" (22.86mm)	1.1" (27.94mm)
Clock Speed	80 MHz	80 MHz	80 MHz
USB to Serial	CP2102	CP2102	CH340G
USB Connector	Micro USB	Micro USB	Micro USB
Operating Voltage	3.3V	3.3V	3.3V
Input Voltage	4.5V-10V	4.5V-10V	4.5V-10V
Flash Memory/SRAM	4 MB / 64 KB	4 MB / 64 KB	4 MB / 64 KB
Digital I/O Pins	11	11	11
Analog In Pins	1	1	1
ADC Range	0-3.3V	0-3.3V	0-3.3V
UART/SPI/I2C	1 / 1 / 1	1 / 1 / 1	1 / 1 / 1
WiFi Built-In	802.11 b/g/n	802.11 b/g/n	802.11 b/g/n
Temperature Range	-40C - 125C	-40C - 125C	-40C - 125C

/C

NodeMCU Pinout and Functions Explained



NodeMCU Compatibility with Arduino IDE

ARDUINO + NodeMCU

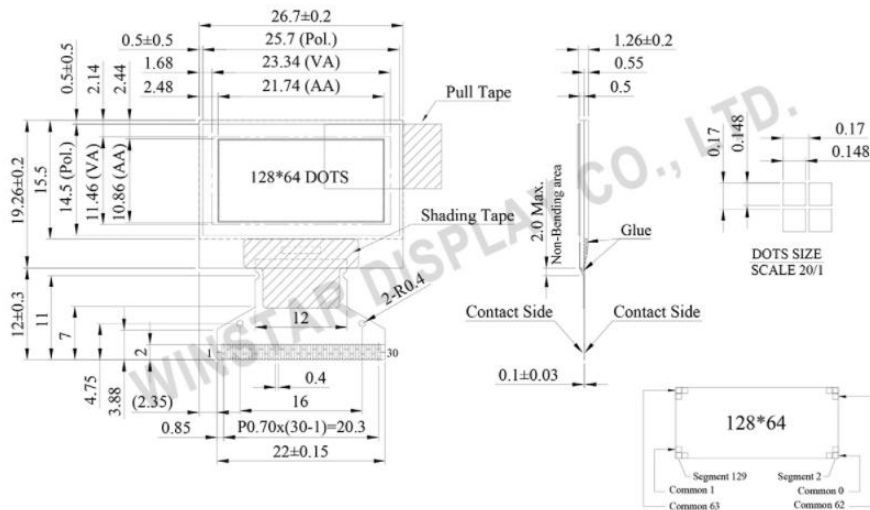


The NodeMCU offers a variety of development environments, including compatibility with the Arduino IDE (Integrated Development Environment). The NodeMCU/ESP8266 community took the IDE selection a step further by creating an Arduino add-on. If you're just getting started programming the ESP8266 or even an established developer, this is the highly recommended environment. Visit our dedicated page on setting up and configuring the Arduino IDE for a NodeMCU ESP8266.



OLED:

OLED display which is made of 128x64 pixels, diagonal size only 0.96 inch. The 128x64 OLED Display has the outline dimension of 26.70 x 19.26 mm and AA size 21.74 x 10.86 mm; it is built in with SH1106 controller IC and it communicates via 6800/8080 8-bit parallel, I2C and 4-wire serial interface. The OLED display which is very thin; lightweight and low power consumption. It is suitable for handheld instruments, meters, mp3, wearable device, IoT, etc. Module can be operating at temperatures from -40°C to $+80^{\circ}\text{C}$; its storage temperatures range from -40°C to $+85^{\circ}\text{C}$.



General Specification

Item	Dimension	Unit
Dot Matrix	128 x 64	-
Module dimension	26.70 × 19.26 × 1.26	mm
Active Area	21.74 × 10.86	mm
Pixel Size	0.148 × 0.148	mm
Pixel Pitch	0.170 × 0.170	mm
Display Mode	Passive Matrix	
Display Color	Monochrome	
Drive Duty	1/64 Duty	
Controller IC	SH1106	
Interface	6800, 8080, SPI, I2C	
Size	0.98 inch	

Absolute Maximum Ratings

Parameter	Symbol	Min	Max	Unit
Supply Voltage for Logic	VDD1	-0.3	3.6	V
Power supply for charge pump circuit	VDD2	-0.3	4.8	V
Supply Voltage for Display	VPP	-0.3	14.5	V
Operating Temperature	TOP	-40	+80	°C
Storage Temperature	TSTG	-40	+85	°C

Electronical Characteristics

DC Electrical Characteristics

Item	Symbol	Condition	Min	Typ	Max	Unit
Supply Voltage for Logic	VDD1	-	2.8	3.0	3.3	V
Power supply for charge pump circuit	VDD2	-	3.5	3.8	4.2	V
Charge Pump Output Voltage for Display (Generated by Internal DC/DC)	Charge Pump VPP	-	7.5	8	8.5	V
High Level Input	VIH	-	0.8×VDD1	-	VDD1	V
Low Level Input	VIL	-	-	-	0.2×VDD1	V
High Level Output	VOH	-	0.8×VDD1	-	VDD1	V
Low Level Output	VOL	-	-	-	0.2×VDD1	V
Operating Current for VDD2 Display 50% ON	IDD2	-	-	10	16	mA

INFRARED SENSOR MODULE:

IR detector is an electronic device, that emits the light in order to smell some object of the surroundings. An IR detector can measure the heat of an object as well as detects the stir. Generally, in the infrared diapason, all the objects radiate some form of thermal radiation.

These types of radiations are unnoticeable to our eyes, but infrared detector can descry these radiations. The emitter is simply an IR LED (Light Emitting Diode) and the sensor is simply an IR photodiode. Photodiode is sensitive to IR light of the same wavelength which is emitted by the IR LED. When IR light falls on the photodiode, the resistances and the affair voltages will change in proportion to the magnitude of the IR light entered. There are five introductory rudiments used in a typical infrared discovery system an infrared source, a transmission medium, optic element, infrared sensors or receivers and signal processing. Infrared spotlights and Infrared LED's of specific wavelength used as infrared sources.

The three main types of media used for infrared transmission are vacuum, atmosphere and optic filaments.

Optic factors are used to concentrate the infrared radiation or to limit the spectral response. Types of IR Sensor There are two types of IR detectors are available and they are, . Active Infrared Sensor Passive Infrared Sensor Active Infrared Detector Active infrared detectors correspond of two rudiments infrared source and infrared sensor. Infrared sources include the LED or infrared ray diode.

Infrared sensors include photodiodes or phototransistors. The energy emitted by the infrared source is reflected by an object and falls on the infrared sensor. Passive Infrared Sensor Passive infrared detectors are principally Infrared sensors. Passive infrared detectors don't use any infrared source and sensor.

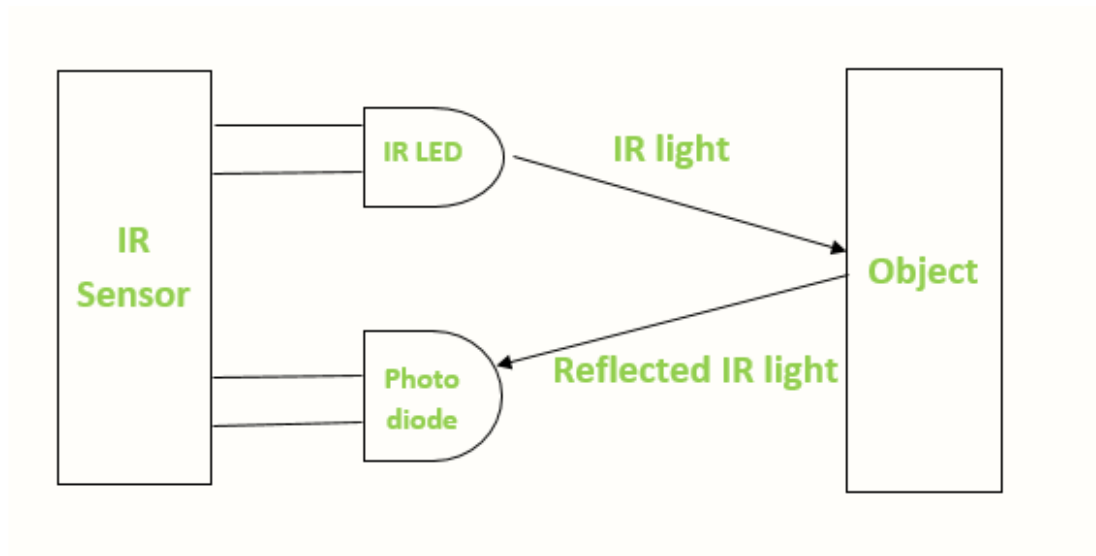
They're of two types amount and thermal. Thermal infrared detectors use infrared energy as the source of heat. Thermocouples, pyroelectric sensors and bolometers are the common types of thermal infrared sensors. Quantum type infrared detectors offer advanced discovery performance. It's faster than thermal type infrared sensors. The print perceptivity of amount type sensors is wavelength dependent. IR Sensor Working Principle There are different types of infrared transmitters depending on their wavelengths, affair power and response time.

An IR detector consists of an IR LED and an IR Photodiode, together they're called as PhotoCoupler or OptoCoupler. IR Transmitter or IR LED Infrared Transmitter is a light emitting diode (LED) which emits infrared radiations called as IR LED's. Indeed though an IR LED looks like a normal LED, the radiation emitted by it's unnoticeable to the mortal eye. The picture of an Infrared LED is shown below. IR LED IR Receiver or Photodiode Infrared receivers or infrared detectors descry the radiation from an IR transmitter. IR receivers come in the form of photodiodes and phototransistors. Infrared Photodiodes are different from normal print diodes as they descry only infrared radiation.

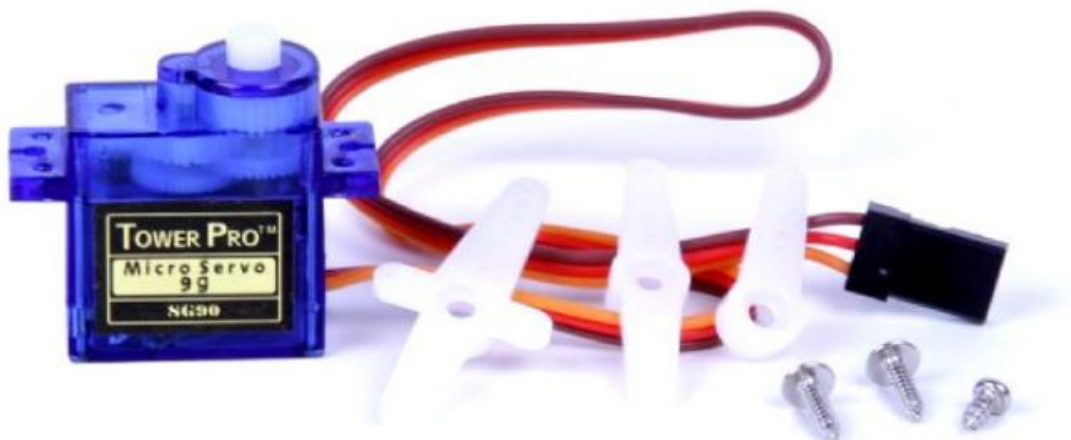
When used in an infrared transmitter – receiver combination, the wavelength of the receiver should match with that of the transmitter. The emitter is an IR LED and the sensor is an IR photodiode. The IR

photodiode is sensitive to the IR light emitted by an IR LED. The photodiode's resistance and output voltage change in proportion to the IR light entered.

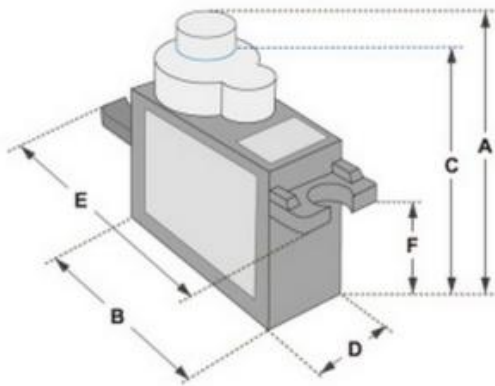
This is the underpinning working principle of the IR detector. IR detector Working When the IR transmitter emits radiation, it reaches the object and some of the radiation reflects back to the IR receiver. Grounded on the intensity of the event by the IR receiver, the output of the detector defines.



9G SERVO:



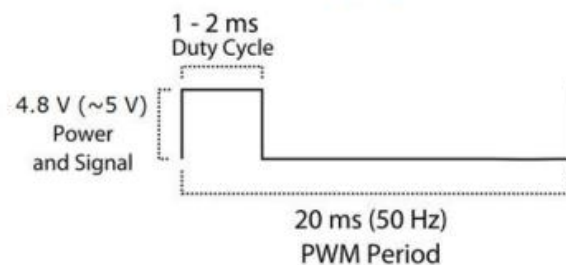
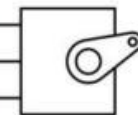
Servo can rotate roughly 180 degrees (90 in each direction), and works just like the standard kinds . but lower. You can use any servo law, tackle or library to control these servos. Good for newcomers who want to make stuff move without erecting a motor regulator with feedback & gear box, especially since it'll fit in small places. It comes with a 3 cornucopias (arms) and tackle.



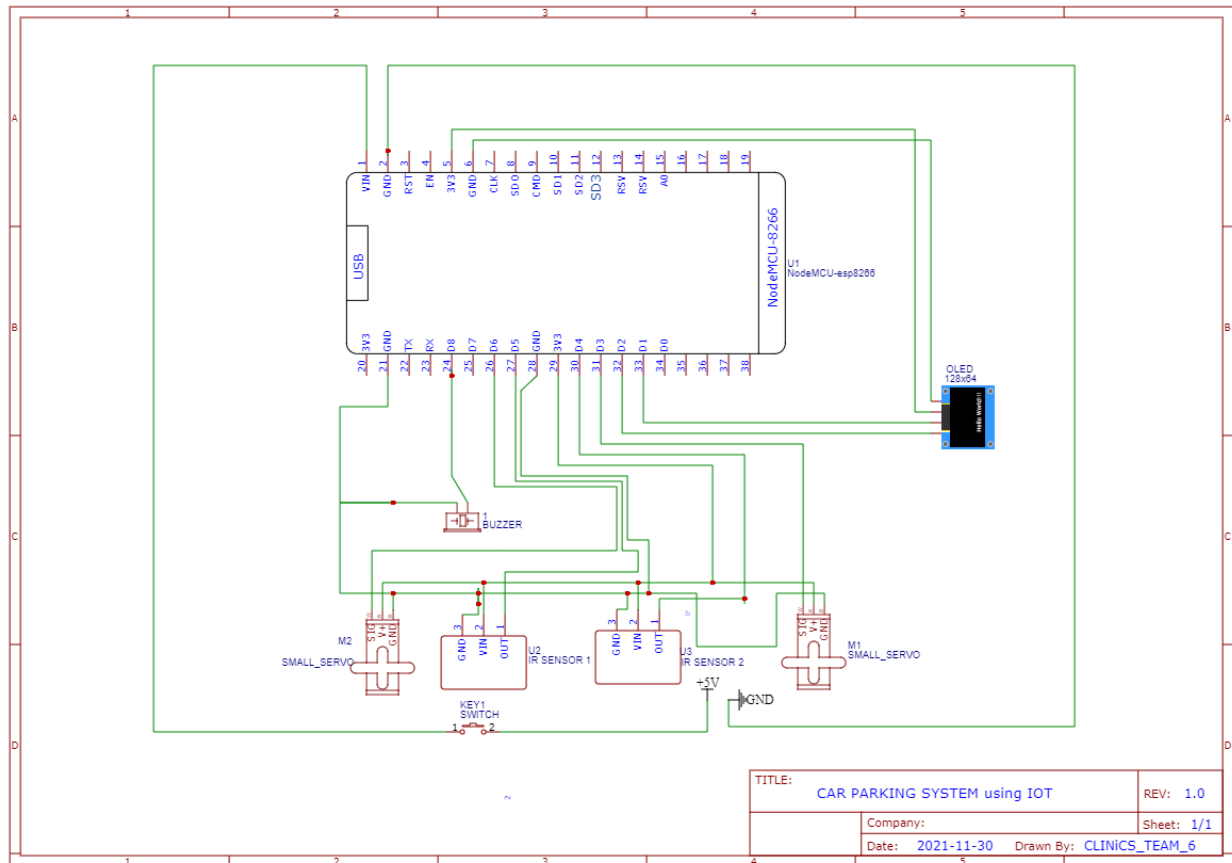
Dimensions & Specifications	
A (mm) :	32
B (mm) :	23
C (mm) :	28.5
D (mm) :	12
E (mm) :	32
F (mm) :	19.5
Speed (sec) :	0.1
Torque (kg-cm) :	2.5
Weight (g) :	14.7
Voltage :	4.8 - 6

Position "0" (1.5 ms pulse) is middle, "90" (~2ms pulse) is middle, is all the way to the right, "-90" (~1ms pulse) is all the way to the left.

PWM=Orange (⏏)
Vcc = Red (+)
Ground=Brown (-)



CIRCUIT SCHEMATIC:



CIRCUIT EXPLANATION:

This circuit is connected to 6v power supply consisting of 4 AA size batteries.

There are two IR sensors used here, one will be at the entrance of the parking area and the other at the exit, the signal pins are connected to D5 and D4 pins of the nodemcu respectively. The power supply to the ir sensors is taken from nodemcu.

Two servo motors are connected and the signal pins are directed to D6 and D3 pins of nodemcu. They are connected to the 6v power supply.

OLED display has four pins SCK, SDA, VCC, GND. It is powered through nodemcu. The SCK pin is connected to D1 and the SDA pin is connected to D2 of nodemcu.

The buzzer is connected to D8 of nodemcu, this signals the working of the circuit by beeping twice.

CODING:

```
#include<ESP8266WiFi.h>
#include<ESP8266WebServer.h>
#include<SPI.h>
#include<Wire.h>
#include<Adafruit_GFX.h>
#include<Adafruit_SSD1306.h>
#include <Servo.h>
```

```
//-----
```

```
ESP8266WebServer server;
Adafruit_SSD1306 display(128,64, &Wire,-1);
const char username[]="CLINICS";
const char password[]="9874563120";
Servo myservo1, myservo2;
int ir_s1 = 14;//connect an ir sensor to pin D5 of nodemcu
int ir_s2 = 2;//connect an ir sensor to pin D4 of nodemcu
int Total = 5;
int Space;
int ocp;
int flag1 = 0;
int flag2 = 0;
```

```
//-----

void setup()
{
  int i;

  //buzzer is connected to D8 pin of nodemcu
  pinMode(15,OUTPUT);

  for(i=0;i<2;i++)
  {
    digitalWrite(15,HIGH);
    delay(50);
    digitalWrite(15,LOW);
    delay(50);
  }

  Serial.begin(9600);
  WiFi.softAP(username,password);
  IPAddress MyIP=WiFi.softAPIP();
  Serial.print(MyIP);
  server.begin();

  pinMode(ir_s1, INPUT);
  pinMode(ir_s2, INPUT);
  myservo1.attach(12);//connect a servo to D6 pin of nodemcu
```

```
myservo1.write(100);

myservo2.attach(0);//connect a servo to D3

myservo2.write(100);


//connect the SCK and SDA pins of OLED to D1 and D2 pins of nodemcu respectively
display.begin(SSD1306_SWITCHCAPVCC, 0x3C);


display.clearDisplay();


display.setTextSize(3);
display.setTextColor(WHITE);
display.setCursor (2,25);
display.println("WELCOME");
display.display();
delay(3000);
display.clearDisplay();


display.setTextSize(2);
display.setTextColor(WHITE);
display.setCursor (46,5);
display.println("CAR");
display.setCursor(24,25);
display.println("PARKING");
display.setCursor (17,45);
display.println(" SYSTEM ");
display.display();
```

```
delay (2000);
```

```
display.clearDisplay();
```

```
Space = Total;
```

```
server.on("/page",page);
```

```
}
```

```
//-----
```

```
void loop()
```

```
{
```

```
server.handleClient();
```

```
if(digitalRead (ir_s1) == LOW )
```

```
{
```

```
if(Space>0)
```

```
{
```

```
myservo1.write(0);
```

```
delay(3000);
```

```
myservo1.write(100);
```

```
Space = Space-1;
```

```
}
```

```
else
```

```
{
```



```
display.clearDisplay();  
display.setTextSize(1);  
display.setTextColor(BLACK,WHITE);  
display.setCursor (22,33);  
display.println("Sorry Slots not");  
display.setCursor(33,41);  
display.println("Available");  
display.display();  
delay (10000);  
}
```

```
}
```

```
if(digitalRead (ir_s2) == LOW )  
{
```

```
myservo2.write(0);  
delay(3000);  
myservo2.write(100);  
if(Space<Total)  
{  
Space = Space+1;  
}  
}
```

```
display.clearDisplay();

display.setTextSize(1);

display.setTextColor(WHITE);

display.setCursor (0,0);

display.println("TOTAL SPACE:");

display.println(Total);

display.setCursor (0,17);

display.println("AVAILABLE SPACE:");

display.println(Space);

display.display();

delay(2000);
```

```
ocp=Total-Space;
```

```
}
```

```
//-----
```

```
void page()
```

```
{
```

```
String mycode=String("No. of parking slots:") + int(Total) + "<br>" + String("No. of parking slots  
available:") + int(Space) + "<br>" + String("No. of parking slots occupied:") + int(ocp) + "<br>";
```

```
mycode= "<html><head><title>nodemcu</title></head><body><body bgcolor=\"white\"><font  
size=\"7\" color=\"orange\"><center><u><b><h1>PARKING SLOTS</h1></b></u></font><b><font  
size=\"6\" color=\"violet\">"+mycode+"</center></font><meta http-equiv=\"refresh\"  
content=\"5\"></body></html>";
```

```
server.send(200,"text/html",mycode);
```

```
}
```

```
//-----
```

WORKING:

The code is uploaded into the nodemcu and the circuit will be ready to operate.

The two IR sensors used here, one will be at the entrance of the parking area and the other at the exit, the signal pins are connected to D5 and D4 pins of the nodemcu respectively. The power supply to the ir sensors is taken from nodemcu. Two servo motors are connected and the signal pins are directed to D6 and D3 pins of nodemcu. They are connected to the 6v power supply. OLED display has four pins SCK, SDA, VCC, GND. It is powered through nodemcu. The SCK pin is connected to D1 and the SDA pin is connected to D2 of nodemcu.

There are 3 global variables declared to calculate the available and occupied space.

First the setup code is run.

The buzzer is connected to D8 of nodemcu.

When the setup code runs the buzzer connected to D8 pin is set to beep twice. The text “WELCOME” and “CAR PARKING SYSTEM” will be displayed in the oled display after the buzzer.

The SSID and Password for the Wi-Fi network is set. And IP address of the Wi-Fi is assigned to a web page. The html code for this webpage to display the information about the parking slots.

The function containing the html code is named as page().

the IR sensors initially sends 5v to the designated pins and the resistance is 0. when an object(CAR) is near the IR sensor the the resistance increases and the voltage becomes 0.

In the loop section of the code is set to work accordingly, the voltage from the IR sensors are repeatedly checked.

The program checks for WiFi clients and allows accesss to the webserver.

When the car is near the IR sensor at the entrance, the voltage through D5 pin is zero and it satisfies the ‘if’ condition , the servo1 is set open and retract to its initial position. After that the available space is calculated by subtracting 1 from the variable Space, each time an object approaches the entrance. When the value stored in Space is 0, the ‘If’ condition becomes false and the barrier will not open.

When all the spaces are occupied “Sorry slots not available”message will be displayed. On the OLED, whenever an object is at the entrance.

When the car leaves the parking area, the IR sensor at the exit, the voltage through D4 pin is zero and it satisfies the 'if' condition, the servo2 is set open and retract to its initial position. After that the space is edited by adding 1 to the variable Space, each time an object approaches the exit. When the value stored in Space is 5, the program stops adding 1 to the Space, but the barrier still opens to allow any vehicle approaching the exit.

This is the working of the program and the circuit.

ADVANTAGES:

- This system allows to park the vehicles with ease.
- It is easy to operate and install.
- Congestion inside a parking area can be avoided.

DISADVANTAGES:

Simultaneous computing takes time when vehicles enter and leave at the same time. So vehicles will have to wait until the barrier opens for them to move.

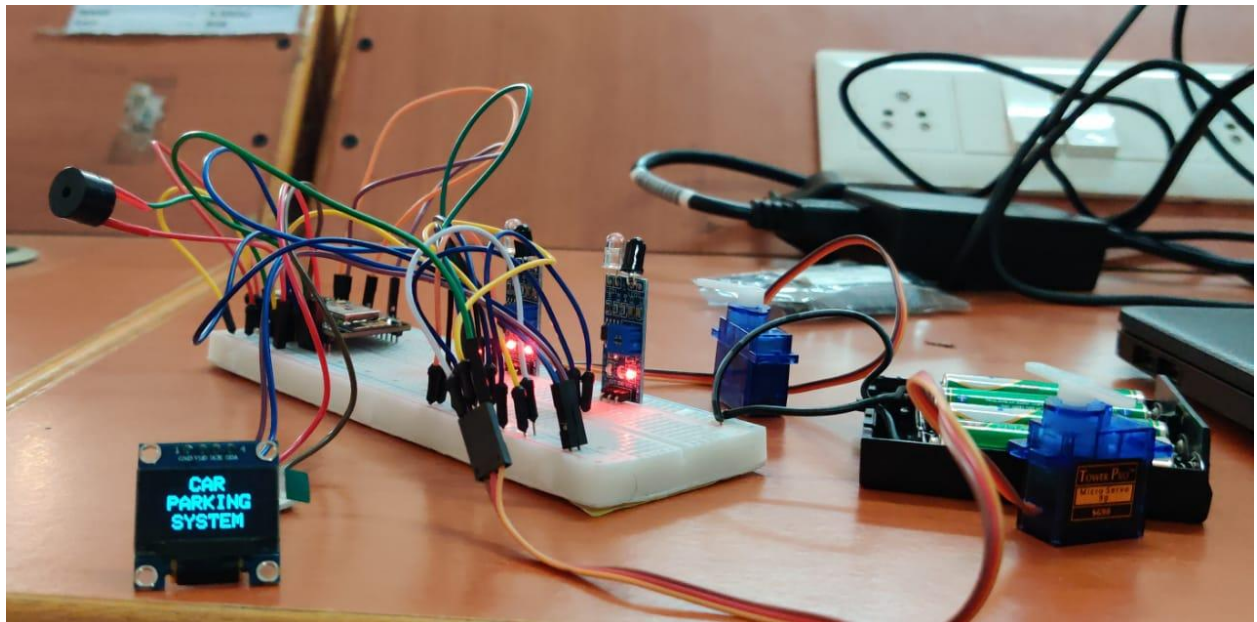
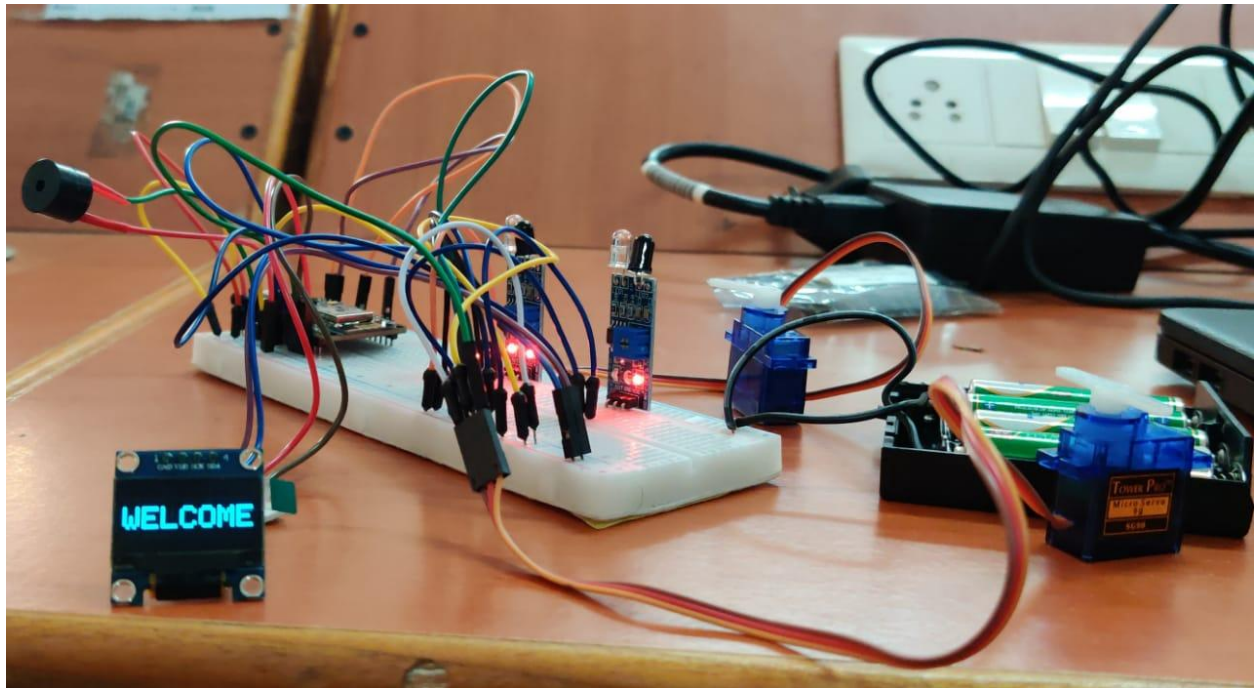
CONCLUSION:

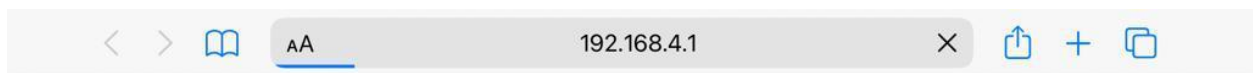
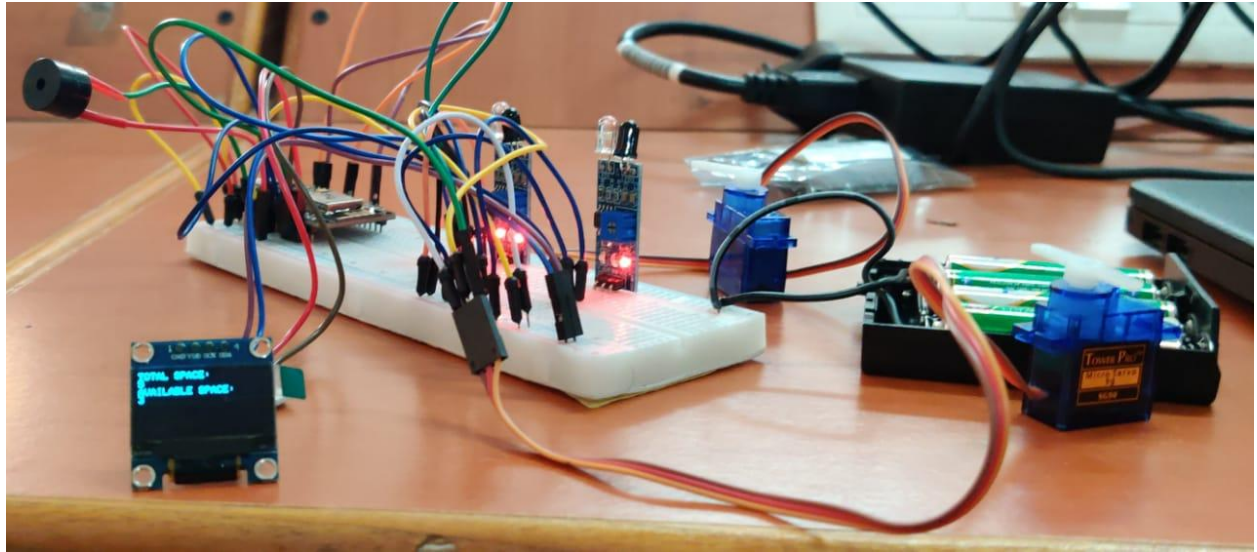
In this modern world the automobile industry is developing exponentially so to ensure safety and avoid problems we developed a system operating systematically while avoiding human errors.

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2. <https://esp8266-shop.com/esp8266-guide/esp8266-nodemcu-pinout/>
3. <https://www.youtube.com/watch?v=DBEBRB0CIMO>
4. [Smart Car Parking System using Arduino Uno. - GitHub](#)
5. <https://github.com/anmspro/Smart-Car-Parking-Syst>
6. [https://iopscience.iop.org/article/pdf/Automated car parking system - IOPscience](https://iopscience.iop.org/article/pdf/Automated%20car%20parking%20system-IOPscience)
7. [\(PDF\) Design and Implementation of Smart Car Parking System](#)
8. https://www.researchgate.net/publication/320356747_...

OUTPUT:





PARKING SLOTS

No. of parking slots:5
No. of parking slots available:5
No. of parking slots occupied:0
